

HARARE, Zimbabwe

WMO: 677750

Lat: 17.930S

Lon: 31.094E

Elev: 4887

StdP: 12.28

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.7	44.6	22.0	19.8	69.4	26.3	24.3	68.9	22.7	59.1	20.2	58.9	4.9	80	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	20.4	88.2	61.3	86.3	61.2	84.5	61.4	69.0	76.9	68.2	76.2	67.4	75.3	9.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.1	120.0	69.8	66.3	116.3	69.6	65.3	112.4	69.4	36.3	77.4	35.6	74.8	35.0	75.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 19.1	16.7	15.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			35.4	71.6	2.3	1.9	33.7	73.0	32.4	74.2	31.1	75.2	29.4	76.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.0	71.1	71.2	70.1	66.9	62.6	58.5	58.1	62.5	68.2	71.4	72.3	71.2
	DBStd	6.09	2.51	2.39	2.60	3.33	3.29	3.40	3.34	4.16	4.53	4.44	3.89	2.68
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	650	1	0	1	19	87	195	214	102	20	7	3	1
	CDD50	6199	654	592	624	506	392	256	253	387	545	664	670	656
	CDD65	1376	190	173	161	75	14	1	1	24	116	206	223	192
	CDH74	9671	876	901	821	503	179	45	49	366	1255	1983	1726	967
	CDH80	2207	138	142	110	47	7	0	1	42	313	665	548	194
(14) Wind	WSAvg	7.4	6.8	6.8	6.9	6.7	6.0	6.3	7.3	7.8	9.1	9.5	8.7	7.2
(15) Precipitation	PrecAvg	30.9	7.8	5.6	3.3	1.5	0.3	0.1	0.1	0.1	0.3	1.5	3.6	6.4
	PrecMax	50.1	18.2	16.5	9.8	4.6	2.6	0.8	1.4	1.1	2.8	4.1	11.5	15.0
	PrecMin	16.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1
	PrecStd	7.5	4.3	3.6	2.4	1.4	0.6	0.2	0.2	0.2	0.6	1.2	2.4	3.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.7	86.1	85.6	84.1	80.8	78.4	78.5	83.9	89.2	91.7	90.9	87.9
		MCWB	64.5	65.1	64.9	62.4	57.7	55.0	53.5	55.5	59.5	61.2	62.1	62.5
	2%	DB	83.4	83.4	82.6	80.9	78.2	75.1	75.1	80.7	86.0	88.6	87.9	84.4
		MCWB	65.1	64.5	64.8	61.8	57.1	54.7	52.6	55.4	58.6	60.7	62.0	63.4
	5%	DB	80.9	81.4	80.8	79.0	75.6	72.3	72.6	78.4	83.9	86.2	85.8	82.1
		MCWB	65.5	64.8	64.9	61.4	56.7	54.2	52.4	54.8	58.0	60.2	61.9	64.1
	10%	DB	79.0	79.4	78.9	77.0	73.7	69.9	69.9	75.2	81.0	84.0	83.5	79.2
		MCWB	65.8	65.0	64.6	60.8	56.4	53.4	52.1	54.0	57.3	59.8	62.0	64.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	69.8	69.8	67.2	62.0	59.2	57.5	59.6	63.0	66.0	67.9	69.3
		MCDB	77.5	78.1	77.2	76.1	71.0	69.8	71.4	75.5	78.9	82.5	78.9	77.2
	2%	WB	68.8	68.4	68.3	65.5	60.3	57.5	55.6	57.9	61.2	64.3	66.5	68.3
		MCDB	76.4	76.5	76.5	75.3	71.5	70.1	69.5	74.7	79.5	80.4	76.8	76.4
	5%	WB	67.9	67.6	67.2	64.5	59.2	56.0	54.3	56.5	60.1	63.1	65.5	67.4
		MCDB	75.6	75.4	75.2	73.4	70.9	68.1	67.6	73.1	78.7	79.1	75.9	75.3
	10%	WB	67.2	66.9	66.3	63.2	58.1	54.7	53.1	55.3	59.0	62.0	64.6	66.6
		MCDB	74.3	74.2	73.8	71.6	69.7	66.1	65.9	71.4	77.1	77.6	75.3	74.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	17.8	18.4	20.7	22.7	23.3	23.3	25.3	25.9	24.1	20.4	17.4
		MCDBR	20.1	20.9	21.0	22.7	25.2	26.3	27.7	28.0	27.2	26.4	24.0	21.6
	5% WB	MCWBR	6.3	6.8	7.0	7.9	9.6	10.3	10.9	9.8	8.9	8.4	7.5	7.0
		MCWBR	16.0	17.1	17.5	19.2	21.4	23.4	23.1	24.4	24.5	22.8	19.0	16.9
(40) Clear-Sky Solar Irradiance	taub	taub	0.337	0.336	0.323	0.289	0.270	0.265	0.276	0.317	0.389	0.385	0.367	0.351
		taud	2.548	2.550	2.593	2.683	2.686	2.656	2.572	2.400	2.220	2.259	2.377	2.491
	Ebn at Noon	Ebn at Noon	319	316	313	313	309	305	303	299	288	297	308	315
		Edh at Noon	35	35	32	27	25	25	28	36	46	46	41	37
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1925	1937	1820	1735	1619	1462	1508	1811	2034	2142	1997	1893
	RadStd	RadStd	146	176	142	106	84	82	84	72	69	116	130	154

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page